WANG Xuewen

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Education

City University of Hong Kong *Hong Kong SAR*

Bachelor of Engineering in Electronic and Electrical Engineering *Sep 2025 - Expected 2029*

- **Affiliation:** Department of Electrical Engineering, College of Engineering.
- **Academic Standing:** GPA 3.65/4.3; Dean's List, Semester B 2025/26.
- **Relevant Coursework:** Introduction to Electronic Design, Foundations of Digital Techniques, Principles of Electrical Engineering; Electronic Devices and Circuits, Computer Programming, Computational Engineering Analysis (in progress).

Technical Skills

Programming Languages: Python, C++, MicroPython.

Hardware & Embedded Systems: Micro:bit, Ultrasonic Sensors, Servo Control, LED Integration, Circuit Wiring.

AI Tools & Workflow: Codex, Prompt Engineering, AI-assisted Development & Debugging.

Prototyping & Validation: Median Filtering, Hysteresis, State-Machine Logic, Software Simulation, Physical Prototyping.

Project Experience

Reliable Smart Parking [GitHub] | *Micro:bit, MicroPython, Python* Feb - Apr 2026

Course Project: GE1354 Introduction to Electronic Design

- Co-developed a Micro:bit-based smart parking system in a four-member team, integrating ultrasonic sensing, servo-controlled access barriers, wireless node communication and OLED status displays.
- Built the physical parking model with teammates and contributed to circuit wiring, MicroPython control logic and proximity-triggered buzzer/LED warnings.
- **Personal extension (Sep - Oct 2026):** Developed hardware-independent Python reliability logic with Codex: five-sample median filtering, 10/12 cm hysteresis and two-reading confirmation; validated state transitions through four passing unit tests and software simulation.
- **Result:** Original team project received the **GE Excellent Student Project Awards: Certificate of Merit**, Semester B 2025/26.

MailPhor [GitHub] | *Python, JavaScript, LLMs* Aug 2026 - Present

- Defined requirements for a local dashboard that turns university emails and digests into structured internship, competition and scholarship cards; built the prototype with Codex.
- Integrated read-only IMAP ingestion and LLM extraction in the prototype, with source references, deadlines and application checklists for reviewing opportunities.
- Refined a bilingual interface with preference-based ranking, bookmarks and persistent application/checklist state through iterative feedback.

EdgeSense Current Monitor [GitHub] | *Python, Filtering, State Machine* Sep 2026

- Built an AI-assisted Python prototype for simulated current monitoring, using median filtering and consecutive-sample rules to identify overcurrent, missing readings and possible sensor freeze.
- Generated fault/recovery logs, CSV evaluation tables and SVG plots to compare false alarms, missed detections and detection delay across reproducible scenarios.

Campus & Volunteer Experience

CityUHK Robocon Team *Sep 2026 - Present*

Hardware Team Member | Student Robotics Team

- Attended hardware induction sessions, learning fundamentals of PCB design, component selection and soldering techniques.
- Gained an understanding of the team's competition robot architecture and collaboration workflows within a multidisciplinary engineering team.

SheNicest Shenzhen Hardware Hackathon *Apr 2026*

Volunteer | Shenzhen

- Handled attendee check-in, on-site guidance and material preparation/distribution during the hardware hackathon.
- Supported participating teams with logistics and physical assembly of small components as requested.

SPIE 思湃义工游学 *Feb - May 2026*

Campus Ambassador

- Used AI tools to help draft promotional copy for volunteering and study-tour programmes, introducing them to university students.
- Published programme articles and posters on WeChat according to the organiser's promotional schedule, completing campus outreach tasks on time.

Languages & Additional Skills

Languages: Mandarin Chinese, Cantonese, English.

Productivity & Collaboration Tools: GitHub, Markdown, LaTeX, Microsoft Excel, PowerPoint, Word.